MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY203 - OPTOELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

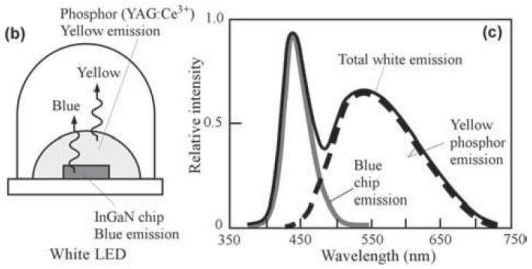
Short Answer

Answer any **10** questions

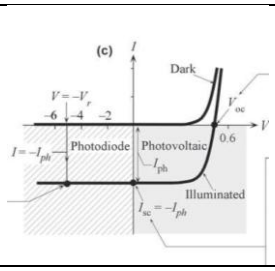
Each question carries 2 marks

1.	What is meant by reverse saturation current?		[K] / [CO1]
Ans.	Reverse saturation current is the small current that flows in a pn junction under reverse bias due to thermally generated minority carriers. It is almost independent of the applied reverse voltage and depends strongly on temperature and material properties. It is denoted by I_0in the diode equation.	Correct definition of reverse saturation current – 2 mark	
2.	Does the mass action law hold true for degenerate semiconductors? Why?		[U] / [CO1]
Ans.	No, the mass action law does not hold for degenerate semiconductors. This is because they are heavily doped and the carrier concentrations no longer follow Maxwell–Boltzmann statistics but instead obey Fermi–Dirac statistics. Hence, the relation $np = n_i^2$is no longer valid.	Correct statement (does not hold for degenerate semiconductors) – 1 mark Reason– 1 mark	
3.	How is a p-type semiconductor formed? Explain with an example.		[U] / [CO1]

Ans.	A p-type semiconductor is formed by doping an intrinsic semiconductor with trivalent impurity atoms, which create holes as the majority charge carriers. These impurities (acceptors) accept electrons and leave behind holes in the valence band. For example, doping silicon with boron (B) produces a p-type semiconductor.	Formation (doping with trivalent/acceptor impurity, holes as majority carriers) – 1 mark Example (e.g., Si doped with boron) – 1 mark	
4.	What is a Phonon?		[U] / [CO1]
Ans.	A phonon is a quantum of lattice vibration in a crystal, representing the quantized energy associated with vibrational motion of atoms.	Definition (quantum of lattice vibration) – 2 mark	
5.	State the expression for the effective density of states in the conduction bandedge?		[U] / [CO1]
Ans.	The effective density of states in the conduction band edge is given by $N_c = 2\left(\frac{2\pi m_e^* k_B T}{h^2}\right)^{3/2}$ where $m_e^*$is the effective mass of electrons, $k_B$is Boltzmann constant, $T$is temperature, and $h$is Planck's constant.	Correct expression for N_c– 1.5 mark Identification of terms (effective mass, temperature dependence, constants) – 0.5 mark	
6.	List any two semiconductor materials used to fabricate LEDs.		[K] / [CO2]
Ans.	Common semiconductor materials used for LEDs include gallium arsenide (GaAs) and gallium phosphide (GaP). These are direct or modified bandgap materials suitable for efficient light emission.	Any two correct materials (e.g., GaAs, GaP, InGaAs, GaN) – 2 marks	
7.	Why are LEDs suitable for optical fiber communication?		[U] / [CO2]
Ans.	LEDs are suitable for optical fiber communication, especially for short-haul applications, because they are simple to drive, cost-effective, and have a long operational lifetime. Although they have a wider spectral width than laser diodes, they provide sufficient output power and work well with graded-index fibers where intermodal dispersion dominates. Hence, LEDs are preferred for short-	-Key advantages (simple, economical, long life) – 1 mark -Suitability for short-haul / graded-index fiber – 1 mark	

	distance, moderate-bandwidth communication systems.		
8.	What is the basic principle of light emission in an LED?		[K] / [CO2]
Ans.	A light-emitting diode (LED) is a pn junction diode fabricated from a direct bandgap semiconductor such as GaAs, in which electron–hole recombination results in the emission of light. The energy of the emitted photon is approximately equal to the bandgap energy, $h\nu \approx E_g$. The basic principle of light emission in an LED is electroluminescence, where electrons and holes recombine in a forward-biased pn junction. This recombination releases energy in the form of photons with energy approximately equal to the bandgap. (The injected electrons recombine with holes in the depletion region and the nearby neutral p-region, known as the active region, producing light through injection electroluminescence.)	Principle (electroluminescence / recombination of electron–hole pairs) – 1 mark Outcome (photon emission ≈ bandgap energy) – 1 mark	
9.	What is the role of phosphor in white LEDs?		[U] / [CO3]
Ans.	In white LEDs, the phosphor absorbs high-energy (blue or UV) light from the LED chip and re-emits lower-energy light (typically yellow) through photoluminescence. The combination of the original blue light and the emitted yellow light produces white light.  	Function of phosphor (absorption and re-emission / photoluminescence) – 2 mark	
10.	What is meant by noise equivalent power (NEP).		[K] / [CO3]

Ans.	The **noise equivalent power** (NEP) is an important property of a photodetector that is frequently quoted. NEP is defined as the required optical input power to achieve a SNR of 1 within a bandwidth of 1 Hz. Stated differently, it is the optical signal power required to generate a photocurrent signal (I_{ph}) that is equal to the total noise current (i_n) in the photodetector at a given wavelength and within a bandwidth of 1 Hz. Suppose that B is the bandwidth of the detector and P_1 is the incident power that results in SNR = 1, then by definition $$\text{NEP} = \frac{\text{Input power for SNR} = 1}{\sqrt{\text{Bandwidth}}} = \frac{P_1}{B^{1/2}} \quad (5.12.5)$$	- Definition– 1 mark - Significance/unit– 1 mark	
11.	What is the basic working principle of a solar cell?		[K] / [CO3]
Ans.	A solar cell works on the photovoltaic effect, where incident photons generate electron–hole pairs in a semiconductor. The built-in electric field of the pn junction separates these carriers, producing a current in the external circuit.	-Principle (photovoltaic effect / EHP generation) – 1 mark - Carrier separation and current generation – 1 mark	
12.	What is the purpose of an antireflection coating on a photodiode?		[U] / [CO3]
Ans.	An antireflection coating is used to reduce the reflection of incident light at the surface of the photodiode. This increases the amount of light entering the semiconductor, thereby improving absorption and quantum efficiency.	Purpose (reduce reflection at surface) – 2 mark	
13.	Define signal-to-noise ratio (SNR).		[K] / [CO3]
Ans.	Signal-to-noise ratio (SNR) is the ratio of the signal power to the noise power in a system. It indicates the quality of the signal, with higher SNR corresponding to better detection and less noise influence.	- Definition (ratio of signal power to noise power) – 2 mark	
14.	What change occurs in the I-V characteristics of a photodiode when illuminated?		[U] / [CO3]
Ans.	When a photodiode is illuminated, a photocurrent is generated which shifts the I–V characteristic downward. This results in a reverse current equal to the photocurrent	-Change in I–V curve (downward shift due to photocurrent) – 1 mark -Result– 1 mark	

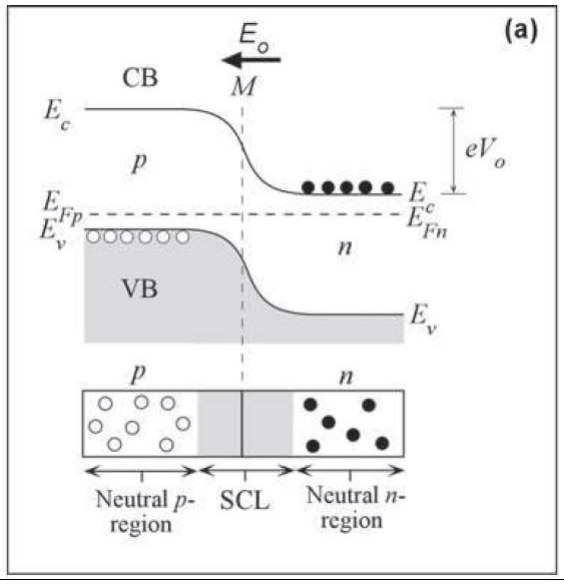
Part B

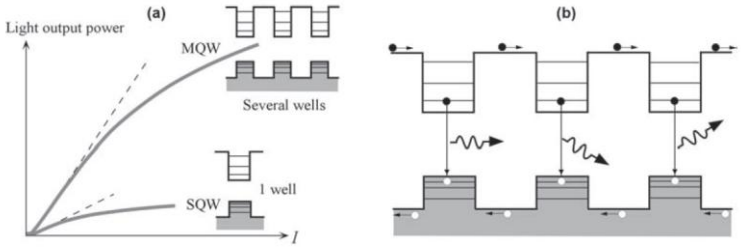
Short Essay

Answer any 6 questions

Each question carries 5 marks

15.	Describe diffusion capacitance and its importance in forward bias operation		[U] / [CO1]
Ans	Page No. 230, Optoelectronics & Photonics, Principles & Practices, Second Edition by S O Kasap Diffusion capacitance arises in a forward-biased pn junction due to the storage of excess minority carriers in the neutral regions. It is defined as the rate of change of stored charge with respect to applied voltage and becomes significant at high forward currents. This capacitance affects the switching speed of the device, making it important in high-frequency applications. takes time; again capacitance-like behavior. The **diffusion** or **storage capacitance** is defined as $C_{\text{diff}} = dQ/dV$. We can easily find C_{diff}. Given $I = Q/\tau_h$ (τ_h being the minority carrier lifetime) under forward bias as in Eq. (3.5.15), we can differentiate Q with respect to V to find *Diffusion capacitance* $$C_{\text{diff}} = \frac{\tau_h I}{V_{\text{th}}} = \frac{\tau_h}{r_d} \quad (3.7.6)$$ It should be mentioned that although under forward bias, C_{dep} can be significant (a few hundred pF mm^{-2}) the diffusion or storage capacitance, C_{diff}, due to the injection minority carriers that are diffusing in the neutral regions far exceeds C_{dep}. In fact, C_{diff} can be several nF mm^{-2}, and a limiting factor in the speed of *pn* junction devices operating in the forward bias regime. We can easily develop a small signal equivalent circuit for a forward-biased junction carrying a current I. There will be a dynamic resistance r_d with two capacitances C_{diff} and C_{dep} in parallel. We should also include a small resistance r_s in series as in Figure 3.24 (c) to represent the resistance of the neutral bulk semiconductor regions. The current has to flow through r_s, which gives rise to a voltage drop through the bulk semiconductor so that not all of applied V drops across the depletion region. Instead of V_o being reduced by V, it is reduced by $V - Ir_s$. It is important to note that Eq. (3.7.6) is grossly simplified. A proper analysis should impose an AC signal of the form $\delta V = V_m \cos(\omega t)$ on top of the DC operating conditions, where ω is the angular frequency of the modulating signal, and derive the signal current δI, and hence the admittance.[13] The final result makes r_d as above in Eq. (3.7.5), but C_{diff} is different by a factor 2, that is, *Diffusion capacitance, long diode (AC)* $$C_{\text{diff}} = \frac{\tau_h I}{2V_{\text{th}}} = \frac{\tau_h}{2r_d} \quad (3.7.7)$$ The lifetime τ_h is the minority carrier lifetime in the lesser doped region (holes in the *n*-side). Further Eq. (3.7.7) is valid when $\omega < 1/\tau_h$. At high frequencies ($\omega > 1/\tau_h$), both r_d and C_{diff} become frequency dependent. For a short diode, r_d remains the same. The diffusion capacitance, however, becomes *Diffusion capacitance, short diode (AC)* $$C_{\text{diff}} = \frac{\tau_t I}{V_{\text{th}}} = \frac{\tau_t}{r_d} \quad (3.7.8)$$ where τ_t is the minority carrier transit (diffusion) time. If we have a short diode as in Figure 3.18, and we decrease the voltage, the excess holes will have to diffuse out. The time it takes for a hole to diffuse across the neutral region width l_n is τ_t, and is given by $\tau_t = l_n^2/2D_h$. For a short diode, $C_{\text{diff}} = \tau_t/r_d$.	-Definition of diffusion capacitance – 3 marks -Importance in forward bias– 2 marks	

16.	Explain the Shockley diode equation and the significance of each term.		[U] / [CO1]
Ans	Page No. 220, Optoelectronics & Photonics, Principles & Practices, Second Edition by S O Kasap	-Correct statement of Shockley equation with the Explanation of parameters – 3 marks - Physical significance – 2 mark	
17.	Explain the formation of a pn junction under open-circuit (equilibrium) condition.		[U] / [CO1]
Ans	Page No. 234, Optoelectronics & Photonics, Principles & Practices, Second Edition by S O Kasap 	-Diagram – 1 marks -Formation of a pn junction under open-circuit – 4 marks	
18.	Compare the internal and external quantum efficiency of an LED		[K] / [CO2]
Ans	Internal quantum efficiency is the ratio of the number of photons generated inside the LED to the number of electron–hole recombinations, indicating how efficiently carriers produce light. External quantum efficiency is the ratio of the number of photons emitted from the LED to the number of injected electrons, and it is lower due to optical losses such as reflection and absorption within the device. Hence, internal efficiency relates to generation, while external efficiency includes both generation and extraction of light.	- Definition of internal quantum efficiency – 2 marks -Definition of external quantum efficiency – 2 marks - Clear distinction	

	Internal quantum efficiency $$\eta_{IQE} = \frac{\text{Rate of radiative recombination}}{\text{Total rate of recombination (radiative and nonradiative)}} \quad (3.14.1)$$ or Internal quantum efficiency $$\eta_{IQE} = \frac{\tau_r^{-1}}{\tau_r^{-1} + \tau_{nr}^{-1}} \quad (3.14.2)$$ The rate of nonradiative recombination rate $(1/\tau_{nr})$ would also include the recombination of injected carriers at the interfaces of heterostructures between different crystals: an important loss mechanism that would reduce η_{IQE}. The total current I is determined by the total rate of recombinations whereas the number of photons emitted per second, the **photon flux** Φ_{ph}, is determined by the rate of radiative recombinations. Internal quantum efficiency $$\eta_{IQE} = \frac{\text{Photons emitted per second}}{\text{Total carriers lost per second}} = \frac{\Phi_{ph}}{I/e} = \frac{P_{o(int)}/h\nu}{I/e} \quad (3.14.3)$$ where $P_{o(int)}$ is the *optical* power generated internally (not yet extracted). The **external quantum efficiency** (EQE) η_{EQE} of an LED represents the efficiency of conversion from electrical quanta, *i.e.*, electrons, that flow into the LED to optical quanta, *i.e.*, photons, that are emitted into the outside world. It incorporates the "internal" efficiency of the radiative recombination process [embedded in Eq. (3.14.1)] and the subsequent efficiency of photon extraction from the device. Suppose that the actual optical power emitted to the ambient, called the **radiant flux**, is P_o. (Φ_e is also used in the literature.) $P_o/h\nu$ is the number of emitted photons per second. Since the number of electrons flowing into the LED is I/e, we have External quantum efficiency $$\eta_{EQE} = \frac{P_o/h\nu}{I/e} \quad (3.14.4)$$ Thus, we should use external QE to meaningfully compare different LED efficiencies and internal QE in comparing different LED materials. For indirect bandgap semiconductors η_{EQE} are generally less than 1% whereas for direct bandgap semiconductors with the right device structure, η_{EQE} can be substantial, for example, 30–40%.	(generation vs extraction losses) – 1 mark	
19.	Explain the structure and operation of a mutiple quantum well LED.		[U] / [CO2]
Ans	Page No. 249, Optoelectronics & Photonics, Principles & Practices, Second Edition by S O Kasap  **FIGURE 3.37** A schematic illustration of the comparison of light power output vs. current characteristics for an SQW and an MQW LED.	-Structure of MQW– 2 marks -operation of a mutiple quantum well LED – 3 marks	
20.	Explain the absorption in direct and indirect band gap materials		[U] / [CO3]
Ans	In direct bandgap semiconductors, photon absorption excites an electron directly from the valence band to the conduction band without change in momentum, leading to strong and efficient absorption. In indirect bandgap semiconductors, absorption requires the assistance of a phonon to conserve momentum, making the process less probable and weaker. Hence, direct materials have a sharp absorption edge, while indirect materials show gradual absorption near the bandgap.	-Diagram-1 marks -Explanation of direct bandgap absorption – 2 marks -Explanation of indirect bandgap absorption (phonon involvement) – 2 marks	

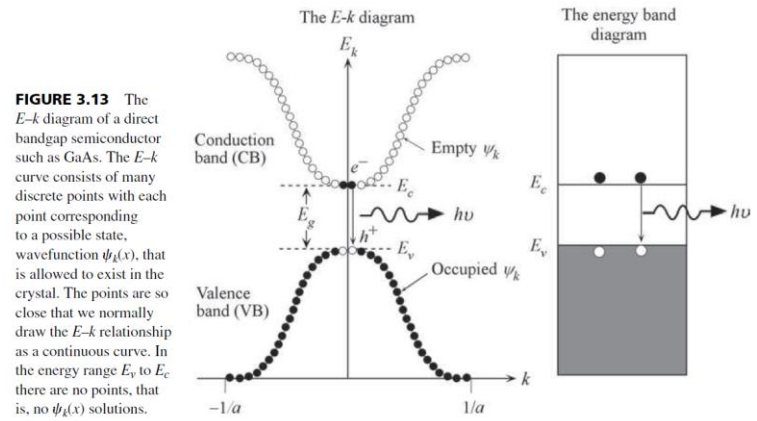


FIGURE 3.13 The *E–k* diagram of a direct bandgap semiconductor such as GaAs. The *E–k* curve consists of many discrete points with each point corresponding to a possible state, wavefunction $\psi_k(x)$, that is allowed to exist in the crystal. The points are so close that we normally draw the *E–k* relationship as a continuous curve. In the energy range E_v to E_c there are no points, that is, no $\psi_k(x)$ solutions.

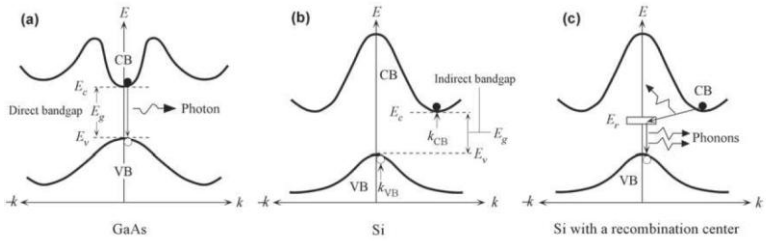


FIGURE 3.14 (a) In GaAs the minimum of the CB is directly above the maximum of the VB. GaAs is therefore a direct bandgap semiconductor. (b) In Si, the minimum of the CB is displaced from the maximum of the VB and Si is an indirect bandgap semiconductor. (c) Recombination of an electron and a hole in Si involves a recombination center.

21.	Explain the processes of carrier generation, drift, and collection in the intrinsic region of a PIN photodiode		[K] / [CO3]
Ans	Page No. 395, Optoelectronics & Photonics, Principles & Practices, Second Edition by S O Kasap	-Diagram – 1 marks -carrier generation, drift, and collection in the intrinsic region-4 marks	
22.	Write a note on multijunction tandem solar cell.		[U] / [CO3]

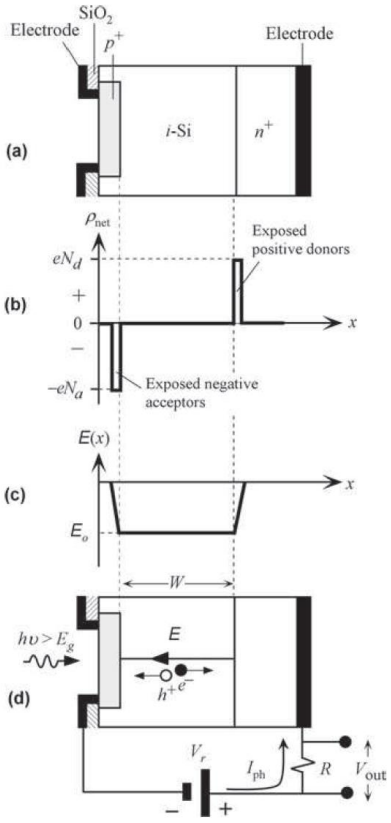
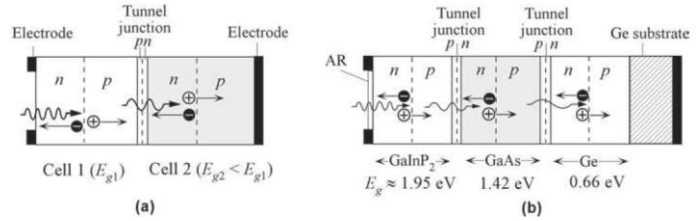


FIGURE 5.9 (a) The schematic structure of an idealized *pin* photodiode. (b) The net space charge density across the photodiode. (c) The built-in field across the diode. (d) The *pin* photodiode reverse biased for photodetection.

Ans	A multijunction tandem solar cell consists of two or more semiconductor layers with different bandgaps stacked together to utilize a wider range of the solar spectrum. The first cell is made from a wider bandgap material and absorbs photons with $h\nu > E_{g1}$, while the second cell absorbs photons that pass through the first layer with $h\nu > E_{g2}$. This layered absorption significantly enhances the overall efficiency compared to single-junction solar cells.  **FIGURE 5.46** (a) A heterojunction solar cell that can absorb both high and low energy photons and generate a photocurrent. (b) A tandem solar cell.	-Structure (multiple layers with different bandgaps) – 1 marks -Working principle with energy condition – 3 marks -Advantage (higher efficiency) – 1 mark	